

Math 432 Implementation Homework 6

Due date: 7 June Sunday

You can get help from AI but write down your own code

1. (30 points) Mount a differential attack to 6-round DES with 1-R technique. Construct a differential characteristic for 5-round DES on your own. Write a program to recover the subkey bits in the sixth round incorporating into the active S-boxes. Write another program for the encryption of plaintext pairs for 6-round DES and collect the data required. Your attack code uses this data for input and recovers the related subkeys in the six round.